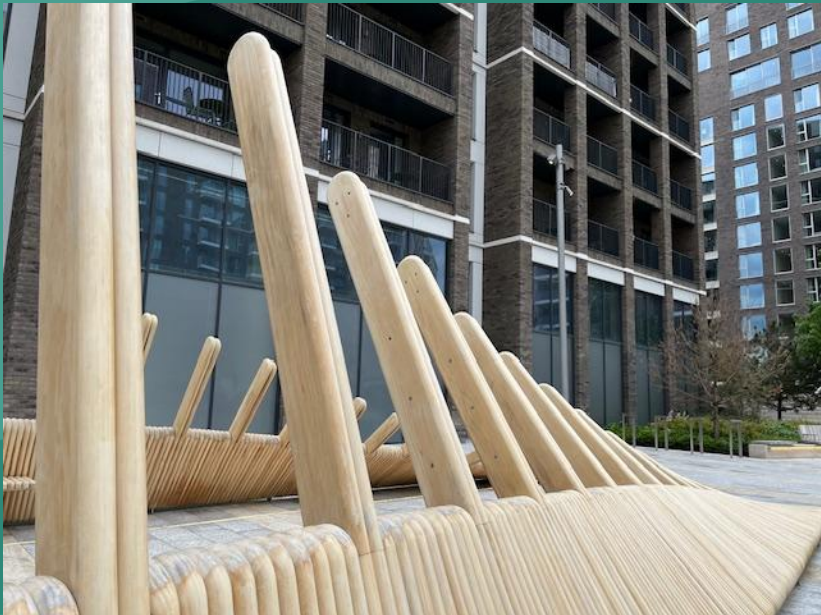


CerPlus® P Series

Nanotechnology Surface Treatment to protect the appearance of Outdoor Timber

General Presentation

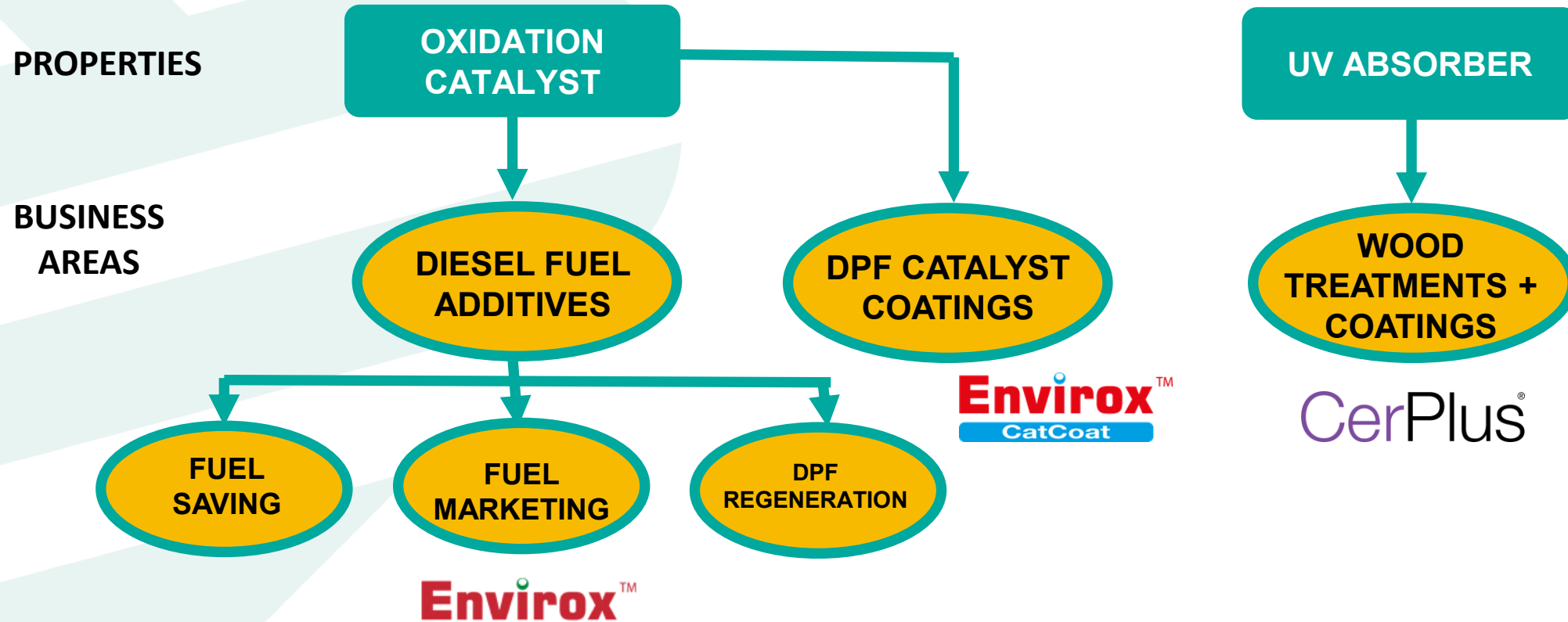


Mike Attfield
Director, Energenics
February 2026

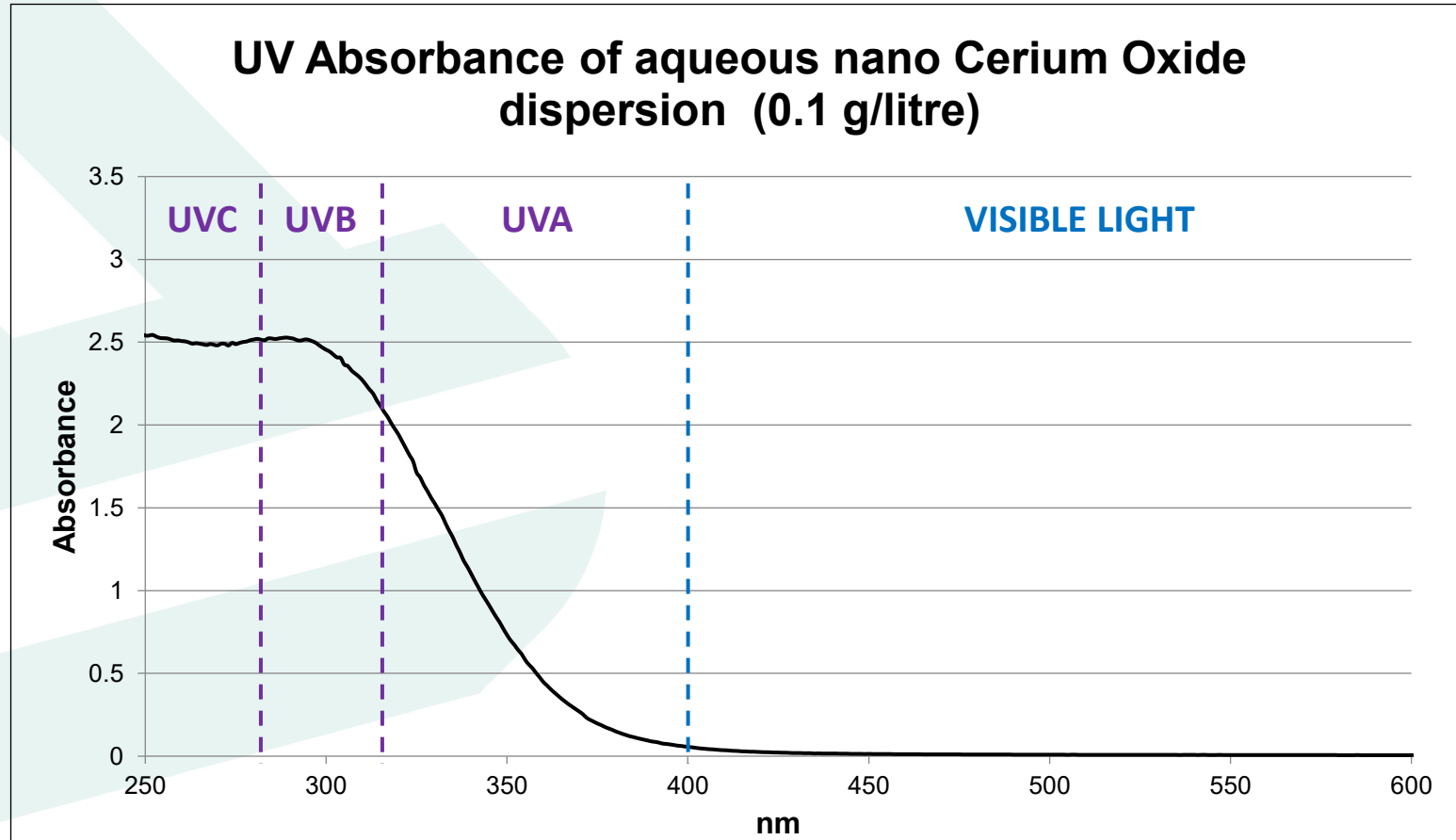
ENERGENICS – COMPANY OVERVIEW



- **NICHE SPECIALITY CHEMICAL COMPANY**
 - Specialist in cerium oxide nanoparticle dispersions
- Oxford University Nanotechnology spin out, established at Begbroke Science Park in 1999 (formerly Oxonica Energy)
- Performance products based on key properties of cerium oxide:



Cerium Oxide – UV ABSORPTION SPECTRUM



- Absorption across all **UVC**, **UVB** and **UVA** wavelengths
- No significant UV Absorption above 400nm (**VISIBLE LIGHT**) = Minimal Coloration

CerPlus® P Series – DIRECT APPLICATION OF WATER BASED CeO₂ NANOPARTICLES TO WOOD

- **MAINTAINS NATURAL BROWN WOOD APPEARANCE** by absorbing harmful UV Radiation
- **WATER REPELLENCY** by making wood surface hydrophobic to protect against water ingress and the resulting biological growth (surface black mould)
- **ENHANCES WOOD COLOUR** by deepening the natural brown tone of darker timbers (not pigmented)



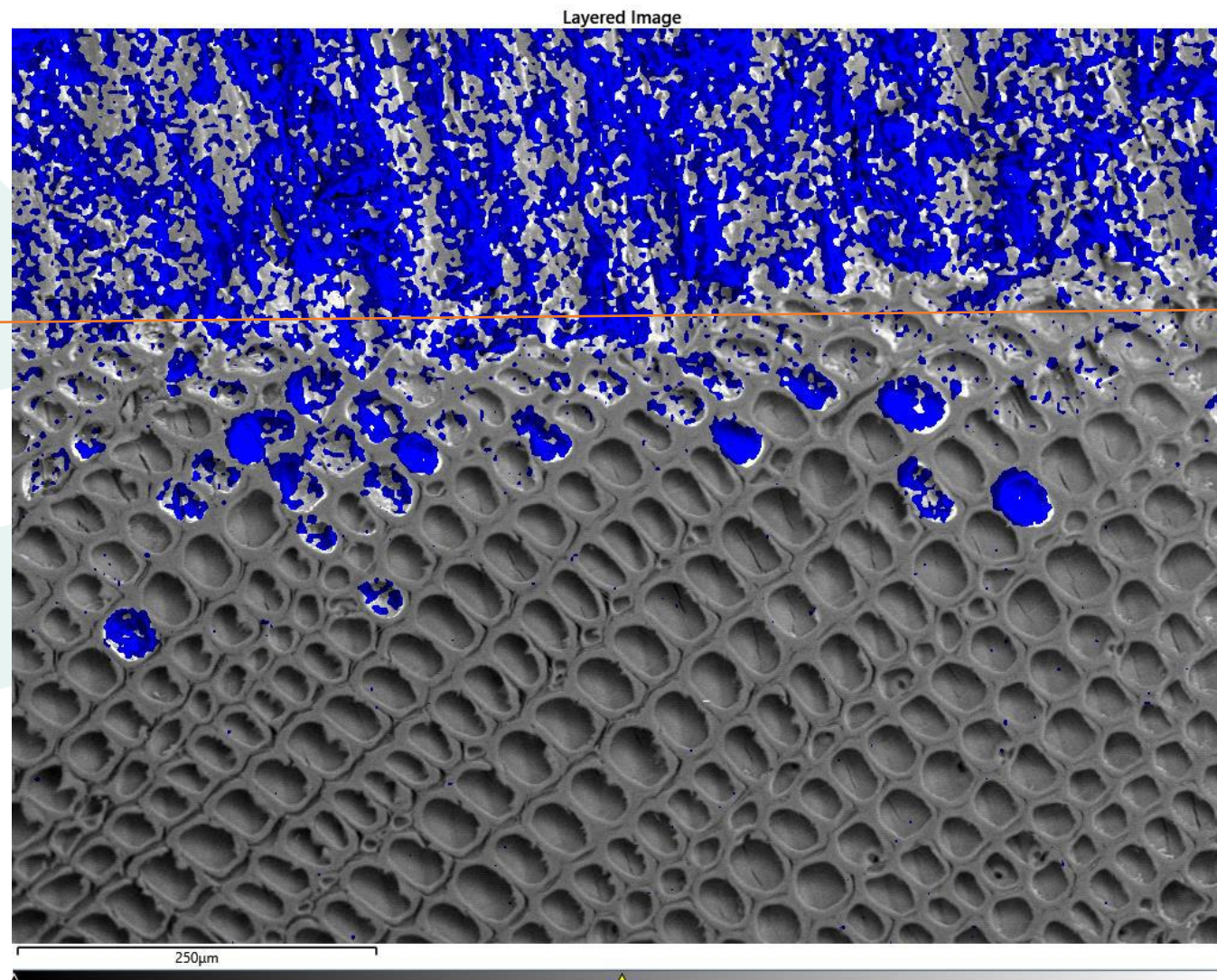
CerPlus[®] P Series – SEM IMAGE (Treated Radiata Pine)

Blue = Cerium

Surface

Vertical
Section

Sub-surface
penetration and
absorption of
active ingredient



CerPlus® P Series – PERFORMANCE TRIAL

Bin Enclosure (Begbroke, UK): 4 Years



CerPlus® P101 PERFORMANCE TRIAL

Red Cedar Garden Sauna (Horsham, UK): 12 Months



Decking Oil
only

(1) CerPlus® P101
(2) Decking Oil

CerPlus[®] P Series – WEATHERING STUDY (Western Red Cedar)

No Treatment (control)

CerPlus P101 (2.5% CeO₂)

CerPlus P103 (5.0% CeO₂)

CerPlus P105 (10.0% CeO₂)



0

6

12

18 months

CerPlus[®] P Series – WEATHERING STUDY (Larch)



No Treatment (control)

CerPlus P101 (2.5% CeO₂)

CerPlus P103 (5.0% CeO₂)

CerPlus P105 (10.0% CeO₂)

0 6 12 18 months

CerPlus[®] P Series – WEATHERING STUDY (European Redwood)



No Treatment (control)

CerPlus P101 (2.5% CeO₂)

CerPlus P103 (5.0% CeO₂)

CerPlus P105 (10.0% CeO₂)

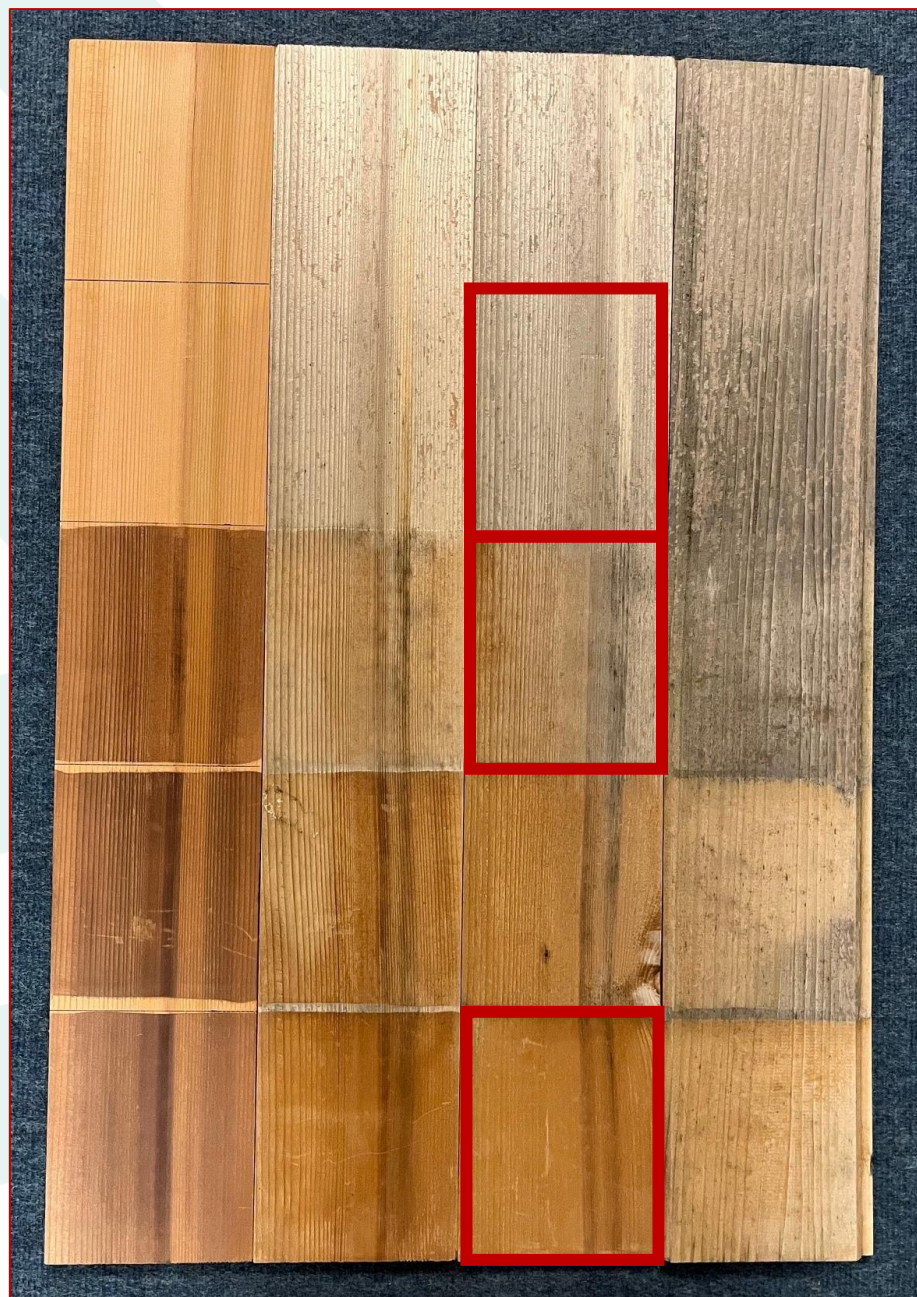
0

6

12

18 months

CerPlus® P Series – **SURFACE ROUGHNESS STUDY** (Western Red Cedar)



0

6

12

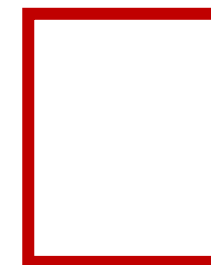
18 months

No Treatment (control)

CerPlus P101 (2.5% CeO₂)

CerPlus P103 (5.0% CeO₂)

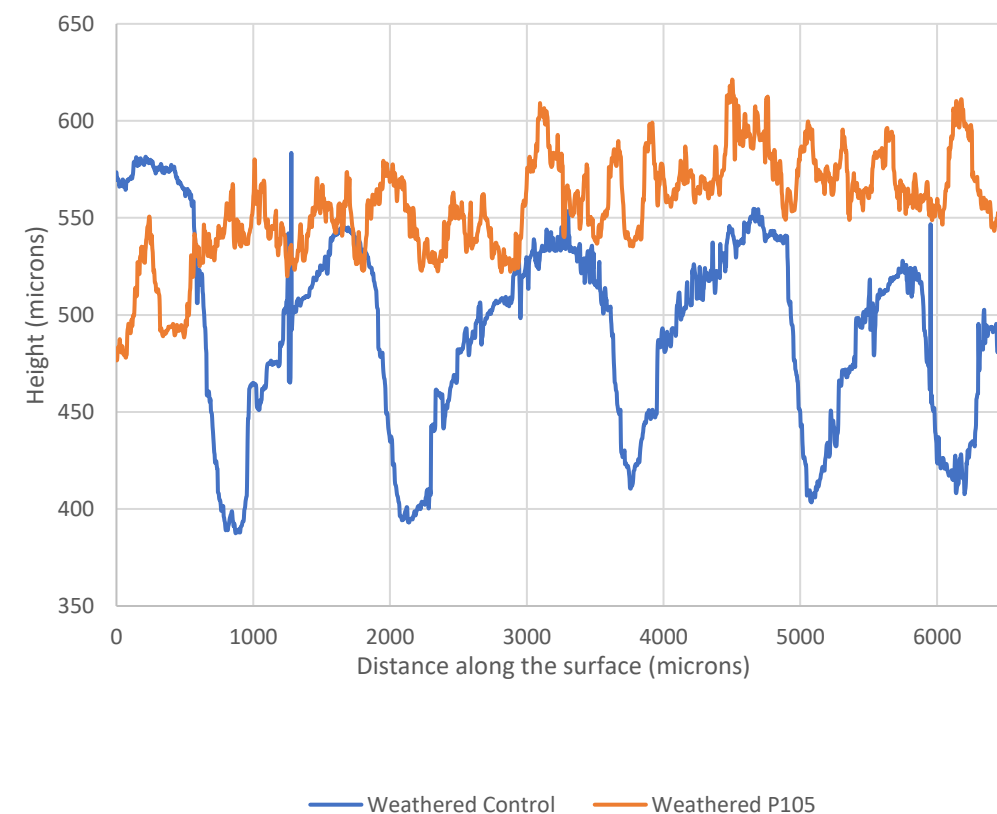
CerPlus P105 (10.0% CeO₂)



**SURFACE
ROUGHNESS
MEASUREMENT
LOCATIONS**

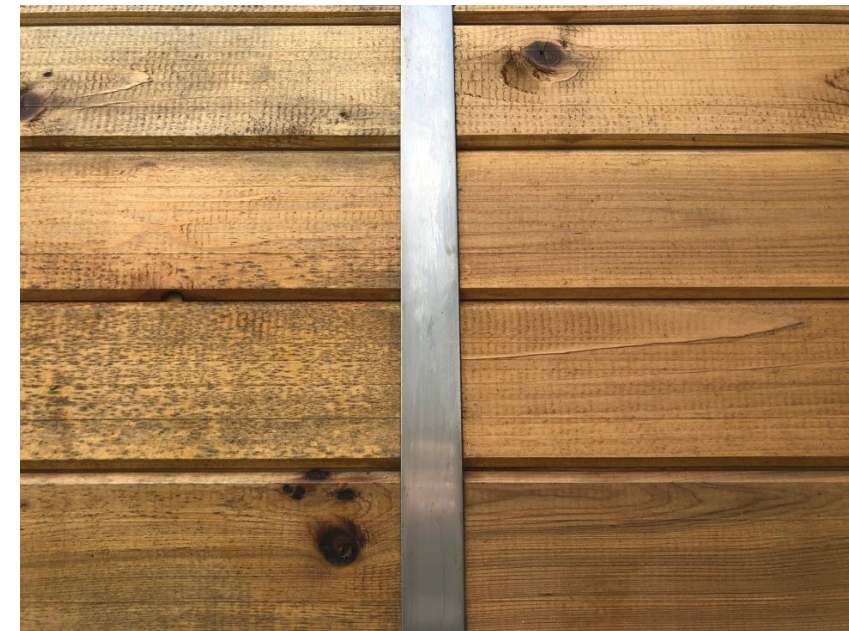
CerPlus® P Series – **SURFACE ROUGHNESS STUDY** (Western Red Cedar, 12 months weathering)

TEST AREA	SURFACE ROUGHNESS (microns)
Unweathered (back of board)	35
Weathered – No Treatment 	210
Weathered – CerPlus® P101 Treatment	46
Weathered - CerPlus® P105 Treatment 	41



CerPlus® P Series – WATER REPELLENCY

- Cerium Oxide Nanoparticles bond to the reactive sites on the surface wood molecules and block them from attack by water
- Wood Surface becomes **HYDROPHOBIC**:
 - Experimental results: Time for disappearance (by absorption or evaporation) of at least 3 x 0.05 ml water droplets placed on wood surface



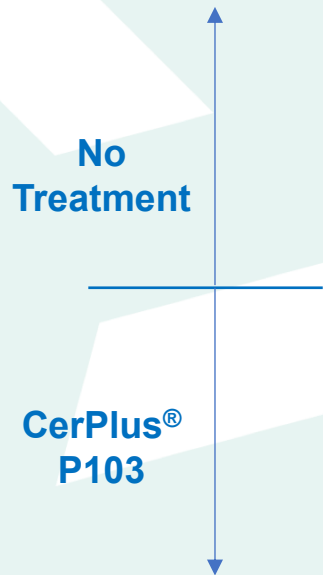
**WESTERN RED
CEDAR GARDEN
SAUNA: 6 months**

TYPE OF WOOD	RADIATA PINE	WESTERN RED CEDAR
UNTREATED (Control)	1-5 mins	3-15 mins
TREATED (Test)	24-64 mins	50-90 mins

CerPlus® P Series – WATER REPELLENCY



HYDROPHOBIC wood surface limits water ingress, inhibiting biological growth (surface black mould)



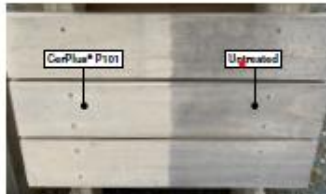
ROBINIA WOODLAND BENCH: 13 months

APPROVED FOR ACCOYA – System Partner



CerPlus® P101 with Accoya® – preserving the natural beauty of new wood

CerPlus® P101 is an innovative smart treatment for protecting the appearance of new wood. It is manufactured by Energenics, a UK performance chemicals business that started as a spin out from Oxford University in the field of nanotechnology.



Test Rig, Accoya Technical Centre, Arnhem, the Netherlands
12 months weathering. South facing cladding panels

Accoya is an ultra high performance, sustainable wood, the material of choice for wood structures where longevity, structural integrity retention and environmental factors are key. It is manufactured by Accoya, the global leader in the manufacture and supply of acetylated wood.

Accoya wood is a preferred material in outdoor applications such as cladding, decking and garden structures and landscaping due to its visual appeal, sustainability attributes, and high performance properties such as durability and stability. Like all wood species, the natural weathering process arising from exposure to sunlight and moisture can lead to rapid changes in appearance.

- Fading and bleaching of the natural, light straw colour by sunlight.
- Surface degradation to a gray colour with a fibrous texture.
- Clusters of unsightly surface mould.

CerPlus® P101

– smart treatment to protect wood appearance

Using its proprietary technology Energenics has developed the innovative CerPlus® P101 smart treatment, providing a scientific breakthrough in helping to preserve the new appearance of exterior wood and reduce maintenance (patent applied). It contains nanoparticles of a natural mineral known to be a highly effective UV absorber. These small particles penetrate deep into the wood surface where they chemically bond to the wood molecules providing long-lasting protection against sunlight. Once attached to reactive sites in the wood they block them from attack by water. The wood surface becomes hydrophobic, greatly reducing the ingress of rainwater and other moisture. As a result the surface growth of biological species like mould/fungi (which are dependent on water) is inhibited.

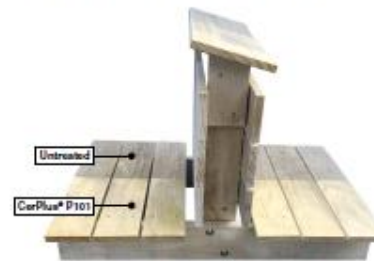
Benefits of CerPlus® P101:

- Provides long-lasting protection against fading and colour changes caused by sunlight.
- Makes wood surface water repellent, inhibiting biological growth (mould/fungi).
- Water-based
- No Biocides
- Minimal VOCs
- Non-Toxic
- Easily applied to the dry surface of new or restored wood using standard techniques like brushing, spraying, dipping or roller coating.
- Can be overlaid with traditional commercial wood coatings, oils or varnishes.



Performance Proven in Partnership – global weathering test programme

Working in close collaboration for over 3 years, technologists from Energenics and Accoya have evaluated and now proven the performance of CerPlus® P101 on Accoya. Using their expertise in outdoor testing, a standardised weathering test rig was designed to give a direct comparison between treated and untreated half panels of new Accoya in cladding and decking orientations and sunny and shaded aspects. Ten identical test rigs were built and located at different sites around the world in Europe, Asia, Australasia, and the USA.



Test Rig, Newquay, Scotland – 12 months weathering

After 12 months exposure a clear picture of enhanced performance emerged:

- CerPlus® P101 maintains the natural light straw colour of new Accoya for significant periods compared to untreated areas, for up to 18 months in some cases.
- CerPlus® P101 performs on all panel orientations but best results are seen on vertical cladding surfaces.
- Transitory dark discolouration due to surface mould & mildew development in more humid climates tends to be excluded on treated areas.
- A smooth surface finish is maintained with no fibre raising.
- When re-applied after cleaning, it maintains the natural light straw colour of Accoya. Surfaces not re-treated after cleaning become silver gray quite rapidly.
- When applied to a weathered silver-gray surface it freezes the position and negates/blows down future graying that normally follows.

Real World Case Study - Switzerland

CerPlus® P101 was applied to new Accoya used in a project for playground equipment in Switzerland. After almost a year the new light straw colour of the Accoya had been retained with minimal evidence of the onset of the transition to silver gray.



CerPlus

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email: info@energenics.co.uk

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United Kingdom
Tel: +44 (0) 1865 233010



CerPlus

CerPlus® P Series – PACKAGE SIZES



- Available in 1 litre jars (clear), 25 litre kegs (blue)
- Surface Coverage = 5 -10 m²/litre

CerPlus® P Series - SMART TREATMENT TO PROTECT WOOD APPEARANCE

- Provides long-lasting protection against fading and colour changes caused by sunlight.
- Makes wood surface water repellent, inhibiting biological growth (black mould, fungi).
- Deepens and enriches the natural brown colour of some woods.
- **Water based formulation**
 - Non-Hazardous
 - No Biocides
- **Easy application** to dry bare wood (new or restored) by brushing, dipping or spraying



THANK YOU



BEFORE (12 months, No Treatment)



AFTER (31 months after P105 Treatment)



CerPlus®